

Research on Driving Factors of COVID-19 Protective Behaviours in Australia

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- Personal protective behaviours played a crucial role.
- Understanding main drivers is important for making public health policy.

Multiple Factors Shape Behaviour

Protective behaviours are influenced by

- Demographic
- Psychological
- Policy factors
- Vaccination
- COVID-19 case numbers
- Other features

Drivers of mask mandates and protective behaviour changes in Australia

- Protective behaviours are not fixed
- The face mask mandates may influence the drivers of protective behaviours in Australia. (Ryan et al. 2025, 8–13)

MLP has shown potential in structured data

- MLP(Multi-Layer Perceptron)s can achieve competitive or even stronger performance on structured data. (Kadra et al. 2021, 9–10)

Research Questions

- How do vaccination rates and COVID-19 case numbers influence the protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional models in predicting protective behaviours?

Imperial College London YouGov Behaviour Tracker dataset

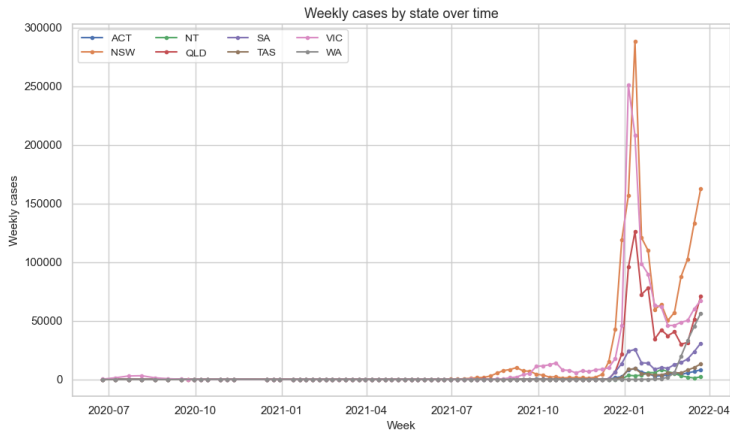
- Demographic and psychological factors (such as gender and age)
- protective behaviours (such as mask wearing and social distancing)
- Comorbidities
- other features

Oxford COVID-19 Government Response Tracker dataset

- The state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023.

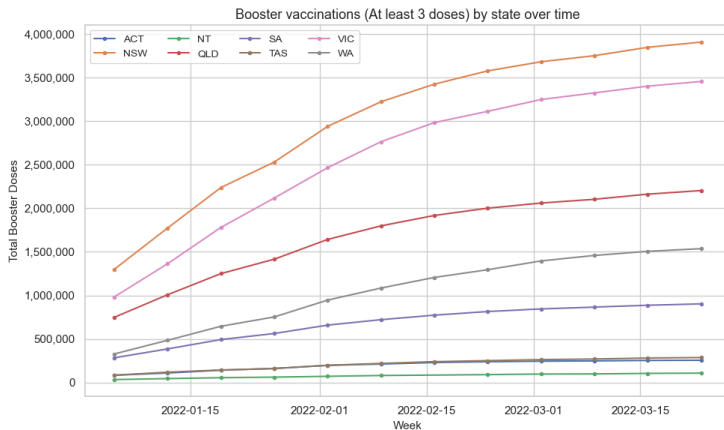
Data-Pandemic Case number

Consolidated dataset for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. (covid19data.com.au 2023a)



Data-Vaccination Data

Vaccine dataset compiled by a third-party team to match the timing and states of the YouGov survey. (covid19data.com.au 2023b)



Government Mandates

- Calculated the daily updated 14-day average of each state's index.
- Rolling average of a certain state reached 3 was considered entered mandate.

Removal of Features with High Missingness

- 10,781 as threshold (Ryan et al. 2025, 3)
- Over threshold were dropped from the dataset.

Repairing Data Gaps Caused by Survey Changes

- YouGov modified the survey between February 10 and October 18, 2021.
- Identified this time window based on dates.
- Filled in with independent 'N/A' categories.

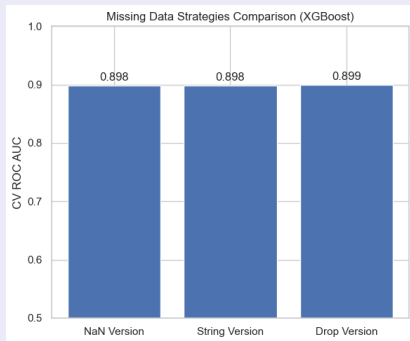
Preprocessing

Experiment on handling structural missing values

Compared three ways on handling structural missing values:

- NAN Version: Retaining np.nan
- string Version: Imputing an “N/A” category
- Drop version: Dropping features directly

Results



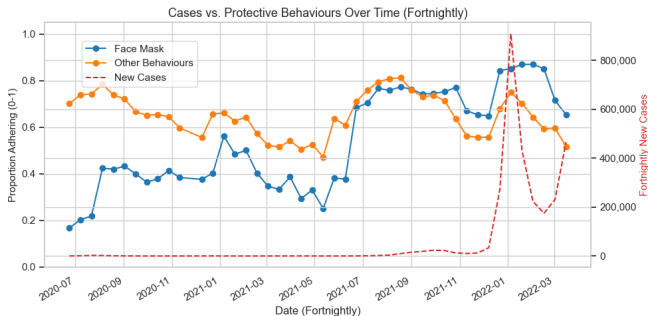
Predictive Variables

Calculated the two-week average and converted to binary using 4 as the cutoff (Ryan et al. 2025, 4) for the following features.

- Mask wearing
- Protective behaviours
- Protective behaviours other than masks.

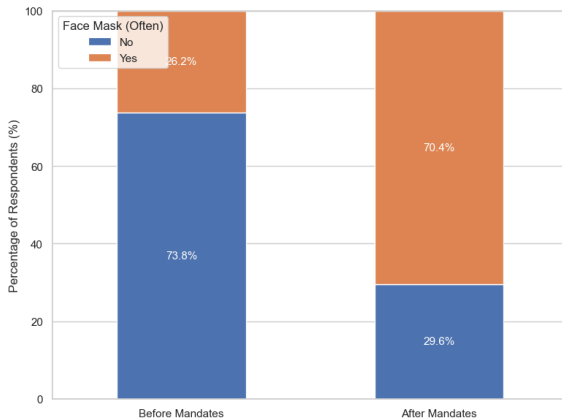
Something interesting

- Protective behaviours track closely with new cases during specific periods.



Model Training Strategy - Overresampling

- This study introduced the random over sampler technique.
- Randomly duplicate minority samples to balance the classes and reduce predictive bias.



Model Training Strategy - Cross Validation

- A single validation set is highly susceptible to the randomness of the initial data split.
- To prevent overfitting during model training, this study implemented a 5-fold cross validation.

How 5-Fold Works

- Split: Divide the dataset into 5 equal parts (folds).
- Train & Test: Train the model on 4 parts, then test on the 1 remaining part.
- Repeat: Rotate 5 times.
- Final Score: Average the 5 results for a robust evaluation.

Model Training Strategy - Evaluation metrics

Main metric

AUC-ROC serves as the primary metric for tuning and model comparison.

Further performance comparison

These metrics offer deeper insights beyond AUC-ROC

- Precision
- Recall
- F1 score

Logistic Regression

- Regularization strength
- Regularization type

Classification Trees

- Maximum depth of the tree
- Minimum samples per leaf
- Minimum samples to split a node

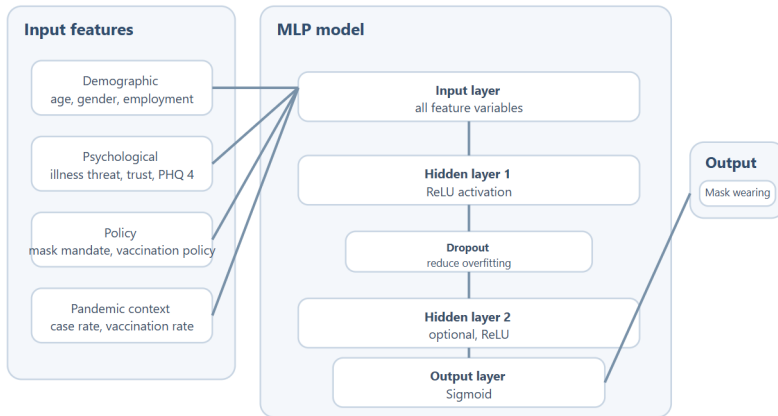
XGBoost

- Learning_rate
- Number of trees
- Maximum depth of each tree
- Number of features per split

Multilayer Perceptron

MODEL STRUCTURE

Proposed MLP architecture for predicting mask wearing



Training: 5 fold cross validation on the training set

Evaluation: F1 score and ROC AUC on the testing set

Interpretation: SHAP values for the two best performing models

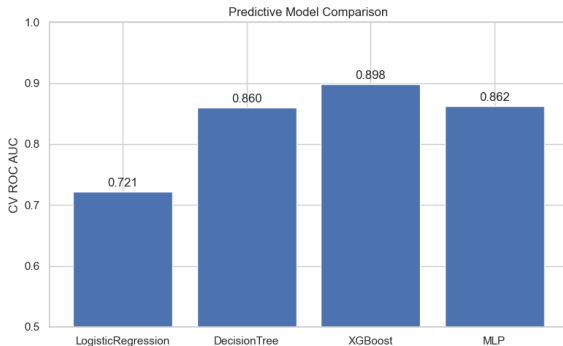
Interpretation

- Method: SHAP (SHapley Additive exPlanations) to the best-performing model.
- Purpose: To open the “black box” and interpret how the model makes decisions.

How SHAP Works

- The “Game”: Treats the model’s prediction as a team game, where each feature is a “player”.
- The Contribution: Calculates exactly how much credit each player deserves for the final score, compared to an average baseline.
- The Impact: Reveals not just the magnitude, but also the direction.

Results- Model Comparison



- Logistic Regression performed the worst.
- MLP performed only slightly better than the Decision Tree.
- The best-performing model was XGBoost.

Influence of Vaccination and COVID-19 Case Numbers

- Investigate feature importance.
- Introduce more detailed data (eg. specific vaccine types & exact locations of new cases).

Deep Learning vs. Traditional Model Performance

- Conduct comprehensive model tuning.
- Expand the scope of the research to include other countries.